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Paleomagnetic constraints on the tectonic evolution of the Sakarya Continent, northwestern Anatolia

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ABSTRACT

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Thirteen sites were sampled in northwestern Anatolia in order to gather paleomagnetic information concerning the tectonic evolution of this region through Jurassic to Early Tertiary time, an interval encompassing much of the history of the Neo-Tethys in this area. Specifically, information concerning the history and behaviour of a crustal block, the Sakarya Continent, was sought in order to place some constraints on geologic models that have been proposed for the region. Four Jurassic sites are believed to exhibit early magnetization signatures that yield similar paleolatitudes for all sites. These results, when compared with paleolatitude information for Eurasia, suggest that Sakarya lay in close proximity to the southern Eurasian margin at least during the Jurassic part of its history. Significantly different declinations from the same four sites are interpreted to reflect the deformation of Sakarya when it docked with Eurasia during the Early Tertiary. These rotations are not attributed to motions of the North Anatolian Fault since results from three Paleocene sites in the same area show no evidence of significant rotation with respect to Eurasia.

Introduction

The origin and evolution of the Neo-Tethyan realm that extends from western Europe to Asia continues to be a significant topic of scientific debate. Part of this debate revolves around tectonic models that have been proposed to explain the complex geologic history of the Pontide Orogenic Belt in Anatolia that now occupies the site of the largely defunct ocean. In order to discriminate one model, or family of models, from another it is often pertinent to focus a part of the debate on some of the details outlined in these models. Such details can be evaluated from a range of perspectives, e.g., structural, stratigraphic or sedimentologic studies. In recent years, paleomagnetic studies of structurally complex zones have been used successfully to enhance our understanding of the tectonic history of such zones (e.g., Van der Voo and Channell, 1980; Kleist et al., 1984; Luyendyk et al, 1985 ; and others). In this paper, we use paleomagnetic methods to evaluate various aspects of the geologic history of the Sakarya Continent, a crustal block identified in the evolutionary model of Sengör and Yilmaz (1981).

Determining the behaviour of the Sakarya Continent from the time of its origin in the Late Triassic up to and including the time of its deformation in the Early Tertiary when the Pontide Orogenic Belt was formed, is the principal focus of this study. Specifically, two goals were identified: (a) establishing the position of the Sakarya Continent relative to Eurasia at different times in the Mesozoic; and (b) establishing whether or not the deformational features identified in this study were principally the product of the Sakarya–Eurasia collision or of the subsequent (Miocene and later) actions of the North Anatolian Fault.

To achieve these goals, rocks of Jurassic, Cretaceous and Early Tertiary age were collected over a broad region of what is believed to be the easternmost portion of the Sakarya Continent.

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The tectonic model

In the model of Sengör and Yilmaz (1981), further elaborated in Sengör et al. (1984), the Sakarya Continent originated during the Late Triassic–Early Jurassic when the westernmost part of the Cimmerian Continent broke apart during the genesis of the Neo-Tethys. The Neo-Tethys was a wedge-shaped ocean narrowing to the west where its geometry became complicated by the presence of the Apulian platform and the Anatolide–Tauride Platform, as well as other, smaller fragments (Fig. 1). These platforms effectively divided the Western End of the Neo-Tethys into two parts, the northern, Vardar Ocean and the southern branch of the Neo-Tethys (part of which is still extant in the form of the Eastern Mediterranean). The Sakarya Continent is proposed to sit within the Vardar Ocean where throughout most of its history it appears to have been a submerged continental fragment. The stratigraphic record of the rocks deposited on this block indicate an early history of shallow-water deposition marked by emergent events indicated by unconformities. For example, the Liassic Bayirkoy Formation is separated from the overlying Dogger–Malm Bilecik Limestone by a disconformity (Ereskay 1965), although Gurpinar (1977) suggests this boundary to be paraconformable. There appear to have been bathymetric variations within the larger platform of Sakarya, as evidenced by the presence of the Mudurnu Basin, a possibly localized depression in which deeper-water facies were deposited (Saner, 1980). This may suggest that Sakarya had a highly variable bathymetric profile and should not be considered as a large, uniform carbonate platform, but rather more like the complex present-day Bahama Platform. Sengör and Yilmaz (1981) also suggest that the Sakarya Continent foundered during the Early Cretaceous, an event marked by the cessation of neritic type deposition and the initiation of pelagic deposition. Throughout the rest of the Cretaceous, it is suggested that the Sakarya Continent remained as a deeply submerged platform until its collision with the southern margin of Laurasia in the Early Tertiary. This collision is marked by the Intra-Pontide suture. Shortly after this Late Paleocene–Early Eocene docking event, the Anatolide–Tauride Platform collided with the southern margin of Sakarya, generating the Izmir–Ankara Suture Zone, and marking the final closure of the Vardar Ocean in this region.

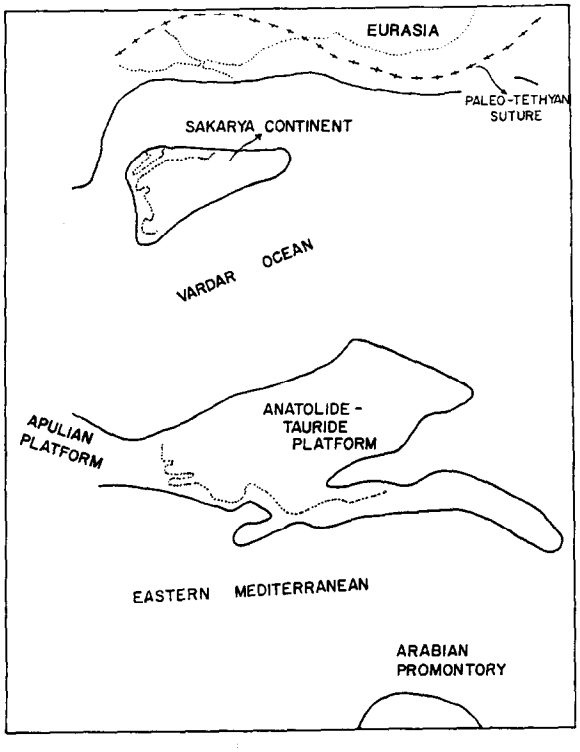


Fig. 1. Paleotectonic map of the Middle Jurassic in the area of modern day Anatolia. Ocean widths not to scale. Fine dotted line indicates outline of modern Turkey and is included for reference only. Modified after Sengör and Yilmaz (1981).

Paleomagnetic study

Sample sites

Thirteen sample sites were occupied in northwestern Anatolia (Fig. 2 and Table 1. At six of the sites, Jurassic age samples were collected, at three of the sites Cretaceous age samples were collected, and at the remaining four sites, Paleocene samples were collected.

Jurassic sites. The Jurassic units sampled were all limestones that ranged in age from Medial to Late Jurassic. These limestones were deposited in a number of environmental settings that include platform, platform margin and deep-water environments. All of the units are the approximate

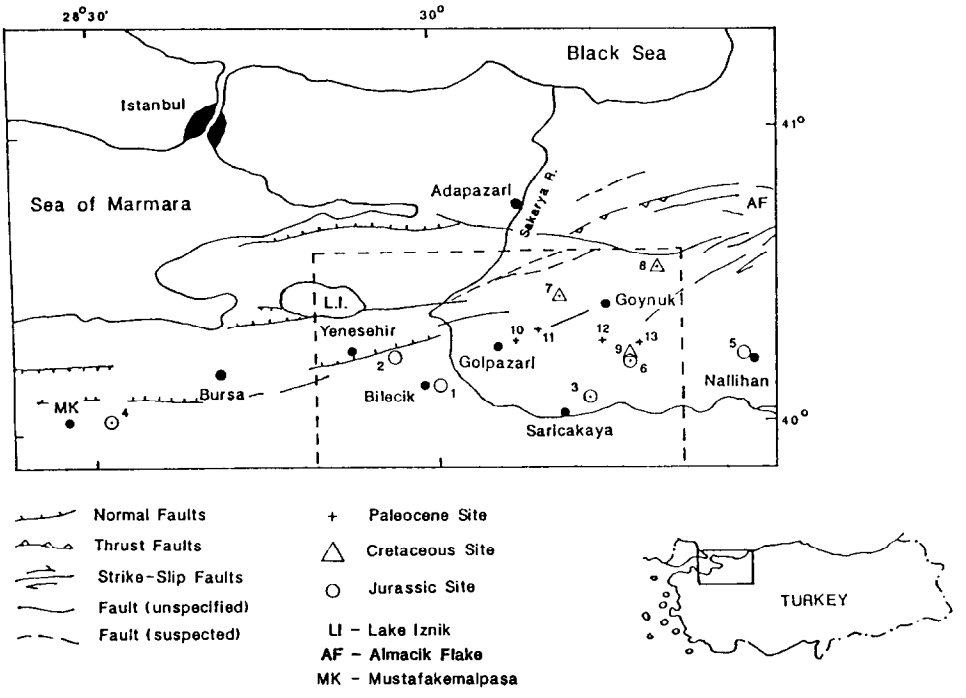


Fig. 2. Map of northwestern Anatolia. Sample sites indicated by number (*1–13*) and by age (as indicated by symbols). Rectangular area outlined by dashed line is enlarged in Fig. 9.

age equivalent of the widespread, platform carbonate unit, the Bilecik Limestone (Ereskay, 1965). This unit is Callovian–Oxfordian in age in the area of the Lithostratotype near the town of Bilecik.

Cretaceous sites. Two of the Cretaceous units

TABLE 1

Sample site information

Age	Site no.	Site location	Number of samples	Dip/Dip direction	Lithology
Jurassic	1	40°09′N, 30°00′E	134	30°SW	micritic-peloidal packstone
	2	40°14′N, 29°49′E	10	50°NW	bioclastic wackestone
	3	40°07′N, 30°39′E	12	26°NW	oolitic grainstone
	4	40°03′N, 28°35′E	12	27°N	pelleted lime mudstone
	5	40°12′N, 31°20′E	12	22°NW	lime mudstone
	6	40°11′N, 30°49′E	12	36°NW	wackestone
Cretaceous	7	40°24′N, 30°31′E	13	32°SW	turbiditic sandstone
	8	40°27′N, 30°57′E	10	18°NW	lime mudstone–wackestone
	9	40°12′N, 30°49′E	12	27°N	lime mudstone
Paleocene	10	40°18′N, 30°22′E	11	6°SE	bioclastic grainstone–packstone
	11	40°20′N, 30°26′E	12	58°SE	medium–coarse-grained quartz sandstone
	12	40°15′N, 30°43′E	12	48°NW	fine–medium-grained quartz sandstone
	13	40°14′N, 30°52′E	12	39°SE	medium–coarse-grained quartz sandstone

sampled in this study were limestones of early Cretaceous age that were deposited in deep-water, pelagic settings, possibly reflecting the foundering event proposed by Sengör and Yilmaz (1981). The third unit was a Late Cretaceous turbidite.

Paleocene sites. Three of the four Paleocene units used in this study are fluvial red beds of the Danian Kizilcay Formation (Ereskay, 1965), a widespread unit in this part of northwest Anatolia. The fourth Paleocene unit is the Selvipinar Limestone, commonly overlain by the Kizilcay Formation and only sporadically developed throughout the region. The Selvipinar Limestone is a platform deposit.

Methods

The natural remanent magnetization (NRM) of samples from all sites was measured using a two-component, cryogenic magnetometer housed in a shielded room. Initial measurements were made at the paleomagnetic laboratory of the University of Texas (then in Galveston). All subsequent measurements were carried out at the Paleomagnetic Research Laboratory (PRL) at the University of Houston. The ambient magnetic field in the shielded room at both laboratories is approximately 200–300 nT. Additional shielding around the magnetometer further reduces the field in the sensor space to less than 2 nT. Alternating field and thermal demagnetization equipment are also housed in the shielded room and further screened from the Earth's magnetic field such that the cooling space in the oven has a residual field of about 5 nT and the AF coil space, a field of less than 1 nT.

Samples from each site were demagnetized using both thermal and alternating field methods. In nearly all cases, thermal demagnetization proved to be more effective in isolating stable components of remanence although stable components were identified with alternating field demagnetization also. Principal component analyses (Kirschvink, 1980) were carried out for all temperature and AF demagnetization ranges in order to better determine the direction of each component of magnetization. In several cases, notably the Paleocene red bed sites, the intensity and direction of magnetization changed little during demagnetization resulting in small vector differences from one demagnetization step to the next. In such cases, more reliable directions were obtained from the vector end points of individual demagnetization steps than from the principal component analysis.

An attempt was made to determine the dominant magnetic minerals present in the various units studied by performing isothermal remanence (IRM) acquisition experiments on samples from those sites where stable components of remanence were identified. Samples were magnetized in progressively stronger magnetic fields up to a maximum of 700 mT and their magnetization measured. In order to better determine the nature of any low and/or high coercivity minerals present, several samples were subsequently stepwise thermally demagnetized and their magnetization measured after each heating step. To monitor changes in mineralogy due to heating, some samples were also exposed to the same maximum magnetizing field (i.e. 700 mT) after each step and their IRM re-measured before heating to the next step. Any substantial changes in this remagnetized intensity indicated corresponding changes in the magnetic mineralogy produced by heating the sample.

Results

Jurassic sites

Of the six sites examined, only Site 4 (Fig. 2) had samples that were weakly magnetized with respect to the sensitivity of the magnetometer. Because of this weakness, demagnetization of samples from this site did not produce reliable determinations of stable remanence. The NRM intensities from the other sites ranged from 3×10^{-8} emu/g to 5×10^{-7} emu/g with average values lying in the lower part of this range. Although there was some variation in the behaviour of samples from each Jurassic site during demagnetization, most displayed a similar two-component magnetization history. All samples possessed a low temperature (0–300°C) component that corresponds roughly with the present day field direction and a higher temperature, more stable component that appears typically in the 300–550°C

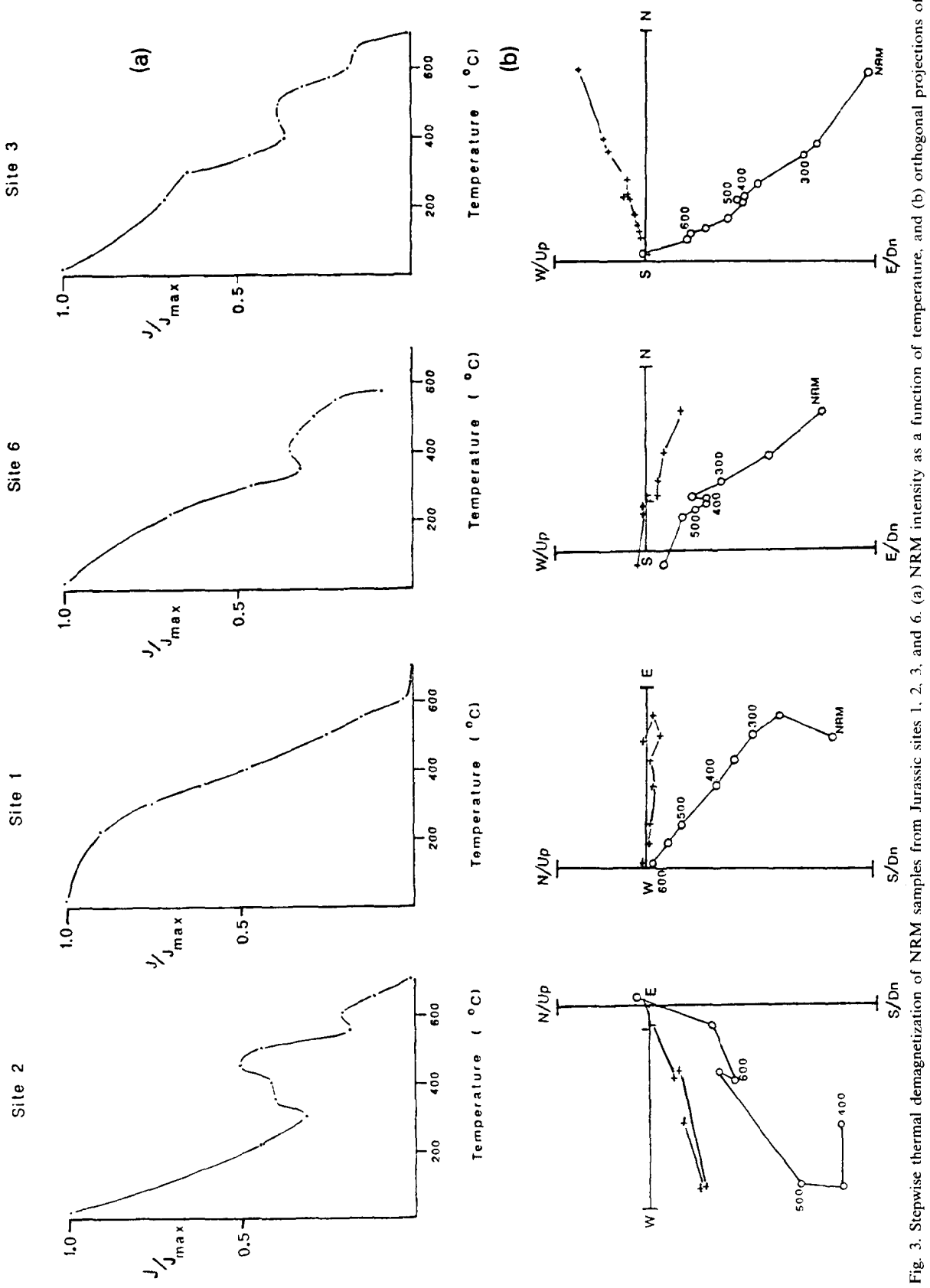


Fig. 3. Stepwise thermal demagnetization of NRM samples from Jurassic sites 1, 2, 3, and 6. (a) NRM intensity as a function of temperature, and (b) orthogonal projections of demagnetization vector end points.

range. Streaking of sample directions between this higher temperature component and the position of the present day field is apparent at most sites. The contribution from the present day overprint, however, varies substantially from site to site with in one case (Site 2) it accounting for 70–80% of the NRM.

Pilot sets of samples from each site were stepwise heated to 700°C and their magnetization measured after each heating step. A majority of samples displayed a significant decrease in intensity between room temperature and 220°C, followed by a much slower decrease through 650°C (Fig. 3). Several samples showed a slight increase between 300°C and 450°C followed again by a decrease through 650°C. Most samples retained sufficient remanence above 575°C to be reliably measured, but heating beyond 650°C frequently produced spurious magnetizations. For Site 6, intensities decreased continuously from room temperature to 575°C with those above 500°C in many cases too weak to reliably measure. Only Site 2 was characterized by dramatic decreases in intensity over the low temperature range indicating the dominance of the present day overprint. Removal of this component resulted in weak, residual magnetizations that were, in several cases, close to the measurement limit of the magnetometer. Consequently it was difficult to isolate higher temperature components in these samples above 400°C. Fortunately, several samples from this site increased in intensity upon heating from 300°C to 450–500°C.

In nearly all cases, the magnetization directions of the stable, higher temperature component for individual sites were consistent at heating steps between 350°C and 500°C. Magnetization directions from Site 5 were different to those of other Jurassic sites in that principal component analysis indicates that all samples carry the same low temperature overprint but have higher temperature components that vary considerably in declination and inclination from sample to sample. None of the magnetization directions from Site 5 matches the high temperature, stable component observed at other sites.

Pilot sets of samples were also demagnetized using alternating fields through 99.9 mT, the maximum possible with the equipment available at the PRL. Intensities tended to decrease rapidly through 15–20 mT then decrease relatively slowly from 20 to 60 mT. Some samples showed an increase in magnetization above 40 mT which was attributed to a anhysteretic magnetization acquired in the demagnetizer. Vector plots of the AF demagnetization also indicate two components of magnetization present in most samples: a low field component that is removed by fields as low as 2.5 mT and a second component that is isolated with fields between 10 and 20 mT. Beyond 20 mT, many of the samples become too weak or too unstable to identify further components. The overall direction of the low field component is approximately along the present field direction. The directions of the 10–20 mT component are very similar to the high temperature component isolated by thermal demagnetization. Spurious results were obtained from AF demagnetization of Site 5 samples with directions at each demagnetizing step forming erratic patterns. Intensity plots of these samples were characterized by several highs and lows that appear random. Only in the low field range (0–5 mT) is there a suggestion of any coherent magnetization.

The carbonate units displayed very similar IRM acquisition characteristics with both low and high coercivity constituent minerals present (Fig. 4). The increase in magnetization between 300 mT and 700 mT is typically 25–30%, but is variable within an individual site. Some Site 2 samples display negligible changes in IRM above 300 mT (Fig. 4). To test for the presence of goethite which has a Neel temperature between 55°C and 130°C depending on crystallinity and isomorphous substitution (Strangway et al., 1968; Hedley, 1971; Bagin et al., 1976), samples were initially heated to 80°C and then to 130°C. Most samples displayed no major or dramatic decrease in magnetization between these temperatures indicating that goethite is not a major constituent magnetic mineral in these samples and is therefore neither the high coercivity mineral identified in the IRM acquisition experiments nor the carrier of the 10–20 mT component of remanence. A few samples from Sites 1 and 2 display a substantial decrease in IRM intensity upon initial heating which can

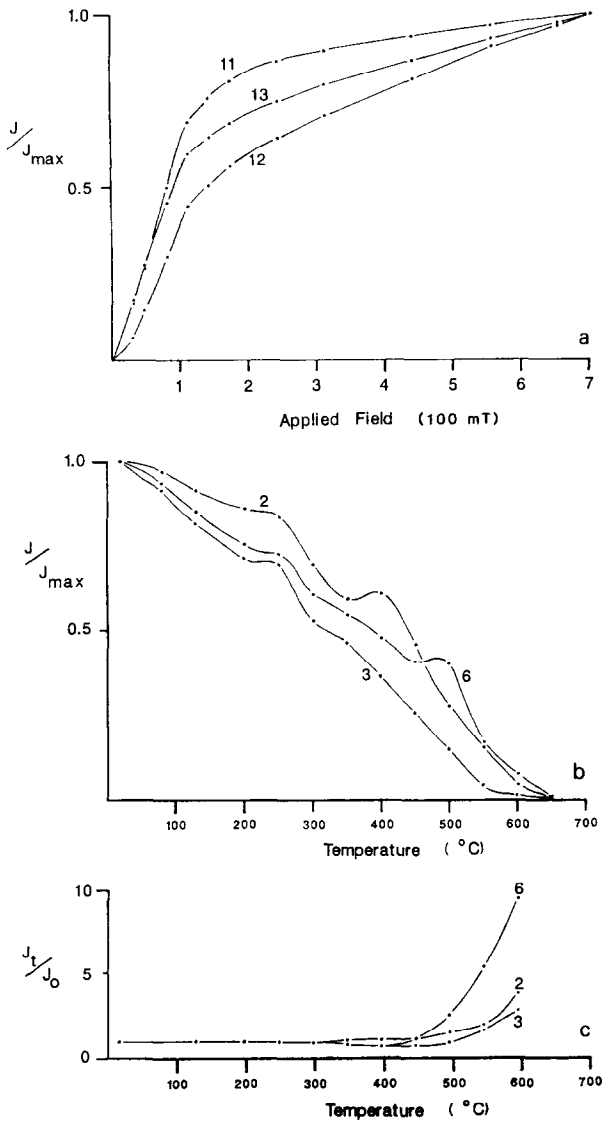


Fig. 4. (a) Isothermal remanence acquisition curves for samples from Jurassic sites 2, 3, and 6. (b) Stepwise thermal demagnetization of IRM acquired in a 700 mT field. (c) Variation in IRM acquired by samples in 700 mT field after each heating step. intensities are normalized to value at 700 mT at room temperature.

be ascribed to the presence of goethite (see Evans et al., 1982). It is likely in these cases that the changes in direction and intensity of their NRM in heating to 220°C is due to the removal of a present day, chemical remanence carried by goethite. In the case of all other samples, it appears more likely that the low temperature component is viscous in origin and is carried by low coercivity grains.

In general, the magnetization of the samples decreases more or less monotonically through 650°C at which step, the remaining magnetization is less than 1% of the initial room temperature IRM. Several demagnetization curves display minor peaks between 400°C and 500°C and slightly steeper slopes in the 500–550°C range, reflecting the presence of a mineral with a blocking temperature in this range (Fig. 4). It is apparent that the higher temperature, stable component of remanence identified in the 300–550°C range during thermal demagnetization, is probably carried by both the low and high coercivity minerals. Samples from Site 6 which are dominated by the low coercivity mineral yield similar paleomagnetic inclinations to those of Site 3 which appears to be more dominated by the high coercivity mineral. Evidence of this can be seen in the results of thermal demagnetization between 550°C and 650°C which shows no perceptible change in magnetization direction of samples from Site 3 (Fig. 3).

The remagnetization intensity after each heating step shows little change over the low temperature range (20–400°C) (Fig. 4). This lack of significant variation suggests that any mineralogical changes are relatively minor and this is viewed as further evidence of the absence of substantial amounts of goethite in the majority of samples. Above 400°C, the remagnetization intensity begins to increase sharply (Fig. 4) In extreme cases, such as Site 6, the intensity increases more than a factor of three or four. This reflects a major change from the high coercivity mineral seen in the IRM acquisition curves to a low coercivity mineral form.

Results from the Jurassic sites, therefore, provided evidence of stable magnetization components in four of the six sites, (viz. Sites 1, 2, 3 and 6). This magnetization is carried by both low and high coercivity minerals which are tentatively identified as magnetite and hematite respectively. Samples from Site 4 were too weakly magnetized to obtain reliable paleomagnetic results, while samples from Site 5 gave spurious magnetizations that though consistent in individual samples, were

nevertheless too widely scattered to produce meaningful site mean directions.

Cretaceous sites

NRM intensities for Sites 8 and 9 were extremely weak (1×10^{-8} emu/g) such that both thermal and AF demagnetization quickly reduced the intensities below the measurement capability of the cryogenic magnetometer. Samples from Site 7 were in general more strongly magnetized with NRM intensities of $2-3 \times 10^{-7}$ emu/g. The low-temperature present field overprint seen at the Jurassic sites was not evident at Site 7. Although all samples decreased in intensity upon heating, none showed any substantial decline in intensity over the 20–300°C range. Intensities, instead, decreased smoothly and continuously over the entire heating range. Intensities remained strong even

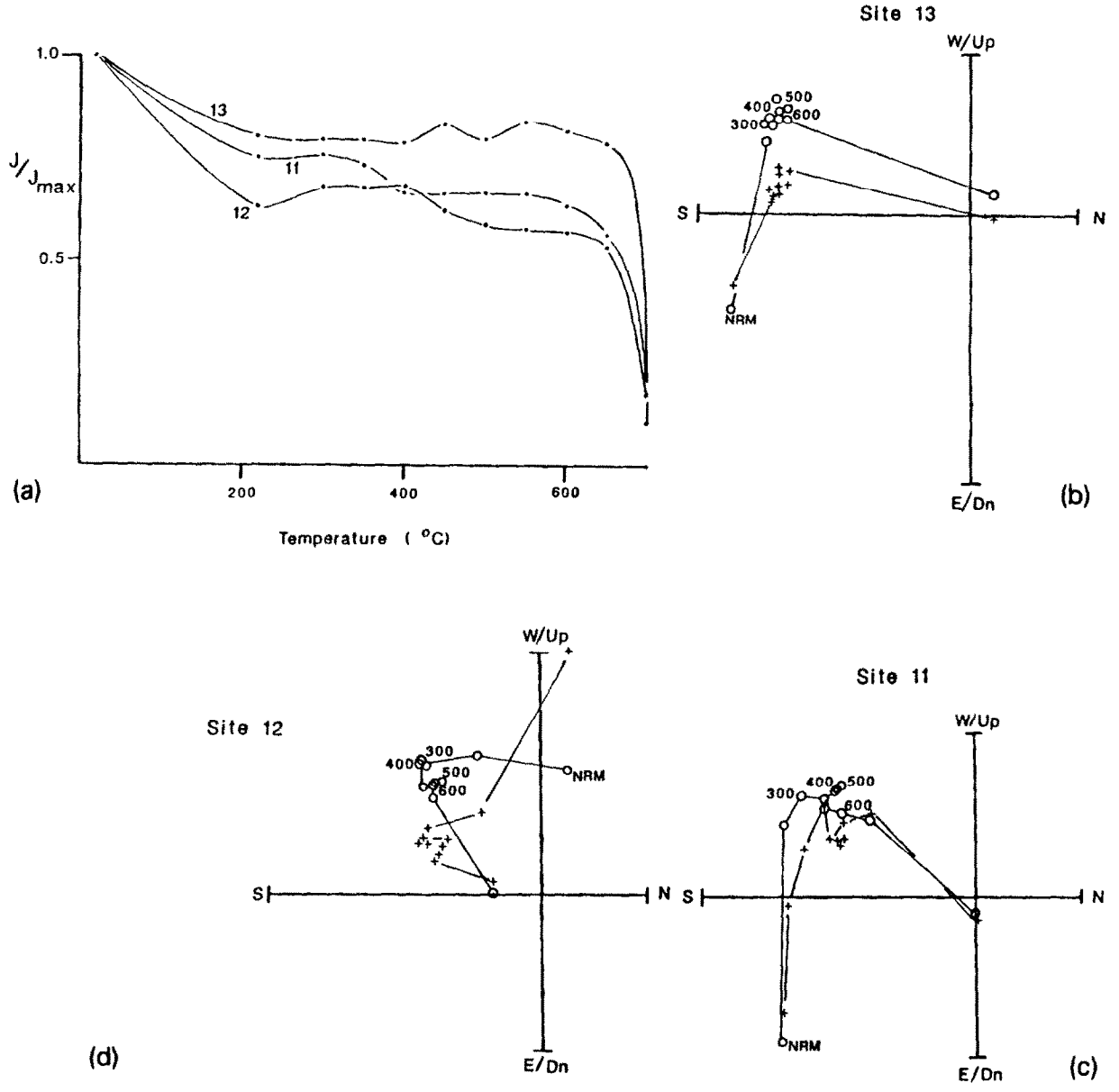


Fig. 5. Stepwise thermal demagnetization of NRM of samples from Paleocene red bed sites 11, 12 and 13. (a) NRM intensity as a function of temperature, and (b)–(d) orthogonal projections of demagnetization vector end points.

above 650°C, although substantial changes in magnetization directions occurred in many cases suggesting a possible remagnetization in the oven. Overall, sample directions between 220°C and 600°C showed considerable scatter and as a result it was not possible to determine a remanence component common to all samples. Similar results were obtained during AF demagnetization, where a reasonably stable component was observed in the 0–5 mT range but above which magnetization directions were spurious and no further stable component could be identified.

Because of the weakness of samples from Sites 8 and 9 and the considerable scatter in magnetization directions at Site 7, no useful paleomagnetic information was obtained from the Cretaceous age rocks.

Paleocene sites

Mean NRM intensities of the red beds (Sites 11, 12 and 13) ranged between 1 and 5×10^{-6} emu/g with samples from Site 12 slightly but not significantly weaker than those from Sites 11 and 13. Intensities of samples from the Selvipinar limestone (Site 10) were by contrast much weaker lying typically in the range $4–8 \times 10^{-8}$ emu/g and were variable.

Thermal demagnetization of samples from the red bed sites produced an initial decrease upon heating to 220°C beyond which the intensities remained fairly constant or decreased very slowly through 600°C (Fig. 5). Site 11 was characterized by a somewhat larger drop in intensity over the low temperature range indicating that these rocks possess a greater contribution from a low temperature component. Above 600°C the intensities of samples from all sites decreased substantially becoming dramatic above 650°C. Above 650°C the directions also became more or less random (Fig. 5). The initial decrease observed in most samples is attributed to the removal of a low temperature, possibly viscous component. Magnetization directions change between room temperature and 300°C and then remain fairly consistent through 600–650°C. As a result, there is a great deal of clustering of points on the demagnetization plots (Fig 5). The site mean directions obtained are essentially constant at all temperature steps between 300°C and 650°C with only modest changes in their statistical reliability. Demagnetization vector plots for Site 11 are similar to those of the other red bed sites in the 220–600°C range. Above 600°C, however, there is a suggestion of a third component present in several samples. This component, seen in approximately half of the samples, is difficult to resolve with the available demagnetization steps but appears to involve slightly steeper and more clockwise directions than those below 600°C.

The behavior of samples from each site differed somewhat during AF demagnetization. In the case of Site 12, there was some small initial change in intensity possibly reflecting the presence of a less stable, low coercivity component but above 2.5 mT, the intensities remain essentially constant through 99.9 mT indicating the presence of high coercivity magnetization components. At Site 13, most samples displayed an appreciable decrease in intensity between 0 and 10 mT which in the case of several samples continued through 20 mT. Between 20 and 40 mT the variation was usually quite modest but above 60 mT, the behavior was very variable with some samples increasing in intensity while others decreased.

At Site 11, intensities varied more noticeably over a wider range of demagnetizing fields decreasing rapidly between 0 and 10 mT and then more slowly through 60 mT. Above 60 mT, several samples showed substantial increases that are believed to be anhysteretic remanences acquired in the demagnetizer.

At all three red bed sites, magnetization directions were easily resolvable into two components: a low field component removed in fields of 2.5 mT or less, and a more stable component that was apparent in the 5–20 mT range. In some cases, this latter component was also present in fields up to 60 mT.

During IRM acquisition experiments samples from each red bed site showed similar rapid increases in magnetization in fields less than 300 mT indicating the presence of low coercivity minerals (Fig. 6). The increase in magnetization continues through 700 mT without any evidence of reaching saturation indicating high coercivity minerals are also present. The increase in magnetization be-

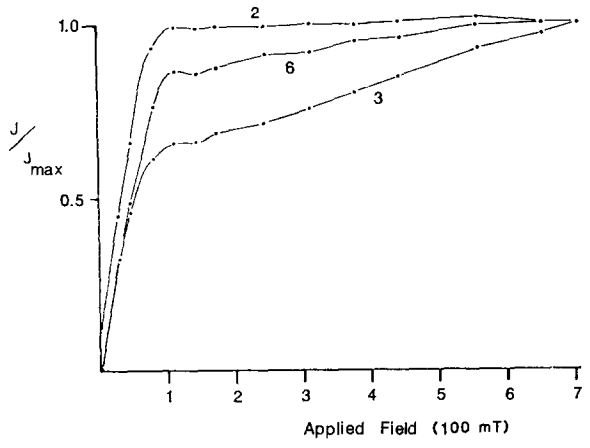


Fig. 6. Isothermal remanence acquisition curves for samples from Paleocene red bed sites 11, 12, and 13. Intensities are normalized to value at 700 mT.

tween 300 and 700 mT lies in the range of 10–30% suggesting that a substantial amount of high coercivity material is present in all the red beds. This high coercivity mineral is hematite, which is abundant throughout the rocks. The low coercivity mineral is probably magnetite but could be maghemite. The IRM samples were not heated and consequently the presence of maghemite could not be confirmed. Thermal demagnetization of the NRM, however, showed no appreciable decrease in intensity between 220 °C and 600°C indicating that maghemite, if present, does not contribute significantly to the stable remanence. Thermal demagnetization of the NRM further indicates that the intensity and direction of stable remanence remains essentially unchanged between 550°C and 650°C. This is consistent with the stable remanence being carried by a mineral with a blocking temperature above the Curie point for magnetite. We therefore conclude that the stable remanence is carried by hematite. The low temperature, present-day overprint observed in all samples from red bed sites is assumed to be viscous in origin and carried by the less coercive grains.

In the case of the limestone unit at Site 10, thermal demagnetization resulted in intensities decreasing by more than 70% on heating from room temperature to 350°C. As a result, above 300°C few samples retained sufficient remanence to be reliably measured. The results of AF demagnetization were similar and consequently only a low temperature overprint was evident in the demagnetization results from this site.

Discussion

Paleomagnetic studies have been carried out in various parts of Turkey (Gregor and Zijderveld, 1964; Van der Voo, 1968; Sanver, 1968; Orbay, 1979; Orbay and Bayburdi, 1979; Lauer, 1981, 1984; Sanver and Ponat, 1981; Evans et al.,1982; Guner, 1982; Saribudak et al., 1989). Only a few of these studies have been in northwestern Anatolia where the remnants of the Sakarya Continent as defined by Sengör and Yilmaz (1981) and Sengör et al. (1984) are located. Paleomagnetic sites on Sakarya include those of Orbay (1979), Lauer (1981), Evans et al. (1982). The recent work of Saribudak et al. (1989) provides evidence from the Eurasian margin immediately north of Sakarya. These studies, which involve Mesozoic and Cenozoic units, can be used to shed light on the structural history of the Sakarya Continent.

A topic of considerable interest to the present study is the relative position of Sakarya with respect to Eurasia through time. Paleolatitudes determined from a Triassic aged flysch site on Sakarya (Lauer, 1981) suggest an equatorial position for the area sometime between the Late Triassic and the end of the Cretaceous. A Jurassic (Kimmeridgian) site near Gerede at the extreme eastern end of Sakarya (Lauer, 1981) gives a structurally corrected paleolatitude of about 4.5 ° N. In situ inclinations are negative implying a significantly more southerly latitude for this site if the remanence was acquired after deformation in the early Tertiary. Cretaceous sites including those of Orbay (1979) and Lauer (1981) give more mixed results. For example, of two upper Cretaceous to Paleocene sites near Gerede (Lauer, 1981) one site gives an inclination of 53.6 ° C while the other has an inclination of only 15.7°. In both cases the units sampled are igneous and no tectonic tilt correction was made. Declinations at these sites also differ substantially. It is apparent from these earlier studies that the paleolatitude of Sakarya in the Late Mesozoic remains unresolved.

In situ directions for the four Jurassic sites in the present study (viz. 1, 2, 3, 6) are widely

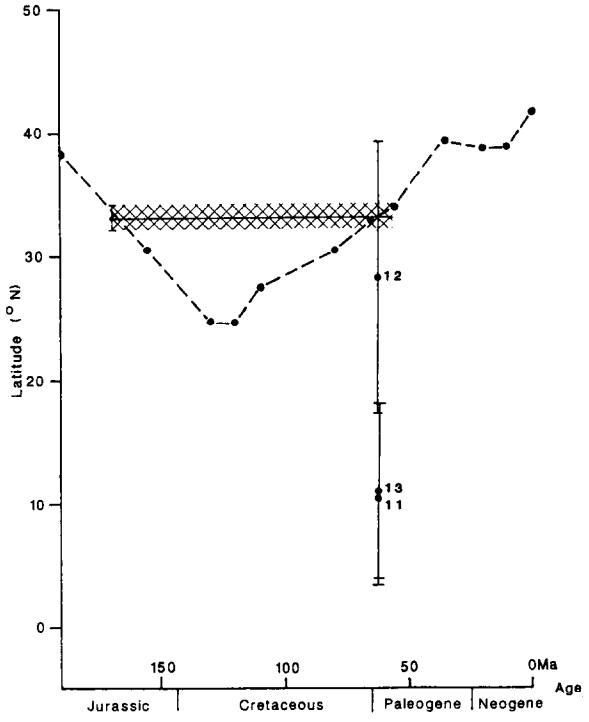


Fig. 7. Variation in latitudinal position of a point on the southern edge of Eurasia from 190 m.y. BP to present (dashed line) based on paleomagnetic poles of Westphal et al. (1986). The present day position of this point is 41.5° N, 30.5° E. Cross-hatched area shows the range of pre-deformation paleolatitudes and age of remanence of Jurassic age sites. Paleolatitudes of Paleocene red bed sites 11, 12, and 13 are shown with respective error bars.

scattered in both inclination and declination (Table 2) with the directions of three of the four corresponding to paleolatitudes much different to those of Eurasia since the time of deformation at the beginning of the Cenozoic (Fig. 7). Site 2 gives a paleolatitude (23–24° N) slightly south of the established APW for Eurasia in the earliest part of the Tertiary. The in-situ declination of this site, however, is S62° W implying a more than 100° counterclockwise rotation with respect to Eurasia. The structurally corrected inclinations on the other hand are remarkably consistent although the declinations still differ significantly. The similarity of the inclination values strongly supports the interpretation that the rocks at each site were magnetized prior to deformation. Although the structurally corrected declinations differ rather markedly from one site to another, the within site variations are generally small (Fig. 8). An exception to this is site 2 where the directions are much more scattered and streak between a westerly direction and the present day field direction. Consequently, we interpret this scatter to the incomplete removal of the present day overprint at this site. As noted, many samples from site 2 are weakly magnetized and during demagnetization frequently become too weak to reliably measure prior to reaching a stable end point. We attribute the

TABLE 2

Site mean remanent magnetization directions *

	In-situ		Tilt corrected		k	α_{95}	n/n_0
	dec.	inc.	dec.	inc.			
Jurassic sites							
1 (thermal)	79.6	26.7	92.8	52.7	3	7.7	134/191
(AF)	76.4	27.0	91.8	50.9	39	6.0	16/16
2	241.7	41.3	299.2	51.3	9	17.2	10/10
3	26.4	70.6	345.5	53.7	391	2.2	12/12
6	290.6	81.2	359.6	52.1	72	6.1	9/12
Paleocene sites							
11	187.3	26.0	185.3	−20.3	13	12.7	12/12
12	265.7	−45.6	207.6	−47.0	16	13.2	9/12
13	174.4	16.6	174.8	−21.2	13	12.6	12/12

* n/n_0 = number of sample characteristic directions used in computing site mean/number of samples demagnetized; k = Fisher's precision parameter.

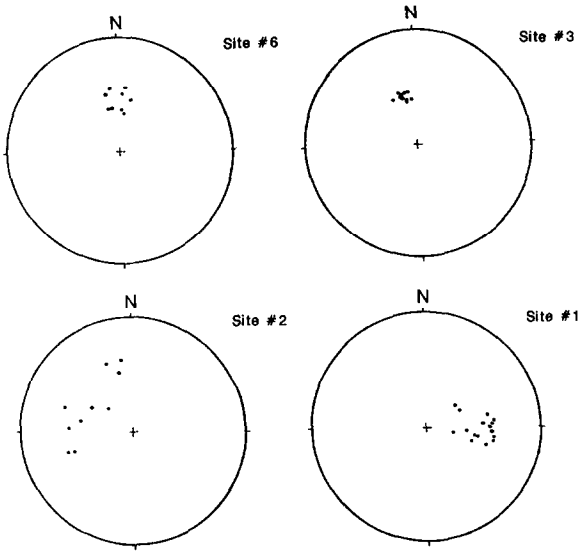


Fig. 8. Stereographic plots of the characteristic directions of stable remanence for Jurassic sites 1, 2, 3, and 6. Dots = lower hemisphere

between-site scatter in declinations to subsequent differential rotations produced by the collision associated with Sakarya docking with southern Eurasia in the Early Tertiary. On this basis, the Jurassic units acquired their magnetization some time between Medial Jurassic when they were deposited and Paleocene when they were deformed. To further test that the correspondence between structurally corrected inclinations is not entirely fortuitous, a statistical study of both in-situ and structurally corrected inclination data was carried out. The analysis was performed using site mean values and, since all sites lie within 1° of one another, all sample values. The results shown in Table 3 clearly indicate the enormous difference in statistical parameters between the in-situ and corrected data for both site means and individual values.

In order to establish a more precise age for the acquisition of remanence, we have tracked the maximum paleolatitude of the study area relative to the southern margin of Eurasia using the pole positions from Westphal et al. (1986) (Fig. 7). These indicate that the Eurasian margin was at, or close to, 40° N in the Early Jurassic and moved south to approximately 25° N by the Early Cretaceous. Subsequently, Eurasia moved north, reaching a paleolatitude of roughly 33° N in the Early Tertiary at the time of the closure of the Vardar Ocean (Sengör and Yilmaz, 1981). Although the uncertainties associated with the Eurasian polar wander path are large, the paleolati-

TABLE 3

Characteristic directions of stable remanence for Jurassic and Paleocene sites

	N	Inclination and declination data				Inclination only data		
		dec.	inc.	k	α_{95}	inc.	k	α_{95}
Jurassic sites								
Site means								
in situ	4	45.6	84.5	3.2	61.8	55.1	11	29.7
tilt corrected	4	357.4	64.9	5.8	42.0	52.0	4282	1.4
All samples								
in-situ	47	63.3	74.6	3.5	13.1	50.4	10	6.9
tilt-corrected	47	12.1	67.2	5.6	9.6	50.4	90	2.2
Paleocene sites								
Site means								
in-situ	3	200.7	0.3	2.1	131.4			
tilt-corrected	3	187.3	−30.1	15.9	32.0			
All samples								
in-situ	33	196.4	4.1	3.1	17.3			
tilt-corrected	33	186.4	−29.7	9.1	8.8			

tudes calculated from the Jurassic sites in this study would suggest that these rocks were magnetized either close to their time of deposition (170–155 Ma) or just prior to the time of deformation in the earliest Tertiary (55–60 Ma) (Fig. 7). Between these times, magnetization events would be expected to produce significantly lower paleolatitudes than those observed. Of these two choices, the early acquisition is supported, in part, by the results from Site 1. At this site, 191 samples collected over a 500 m thick stratigraphic interval (Late Dogger–Early Malm) yielded stratigraphically successive groups of normal and reverse magnetizations that form a magnetic stratigraphy pattern. Significant portions of the lower part of the section were dominated by normal polarity followed by a higher frequency of normal and reverse polarity horizons in the upper part of the section. This pattern is broadly compatible with the record of the geomagnetic field in the Middle to Late Jurassic as outlined by Ogg and Lowrie (1986) and Ogg et al. (1984), but lack of precise biostratigraphic control prevented a more definitive correlation with these established geomagnetic reversal scales. We believe that this magnetostratigraphic record provides supportive evidence of a primary magnetization for this site. The limited stratigraphic range (generally less than 10 m) over which samples were collected at the other Jurassic sites may well explain the lack of such reversal patterns in those cases.

A second topic of interest is the widely scattered declinations reported by several authors (e.g., Lauer, 1981; Evans et al., 1982). It is clear from these values that the remnants of the Sakarya Continent today comprise several, possibly many, small crustal blocks. The scatter in declination values is assumed to have been produced by one or both of the major Cenozoic tectonic events in the region: the early Tertiary collision of Sakarya with Laurasia that closed the Vardar Ocean (Sengör and Yilmaz, 1981; Sengör et al., 1984), and the strike-slip movement along the North Anatolian Fault which began in the Miocene (Sengör, 1979; Sengör et al., 1985). Some authors have linked declination anomalies with motion along the fault (e.g., Van der Voo, 1968; Orbay, 1979). Although several of these studies have been considerably to the east, at least two involve sites close to the study area. Orbay (1979) in his study of Lower Cretaceous rocks near Bolu interprets a 50° counterclockwise rotation to the influence of the fault. Saribudak (pers. commun., 1988) finds similar results near Dokurcun involving the Almacik Flake (Fig. 2) (see also Sengör et al., 1985, especially fig. 14), where he has been able to document a significant clockwise rotation.

To discriminate between the influence of the collision and that of the fault, we have used the results from the three Paleocene red bed sites within the area of Jurassic and Cretaceous anomalous declinations. Sites 11 and 13 have southerly declinations both with and without the tectonic tilt correction (Table 2). In-situ inclinations are positive implying southerly latitudes for these sites whereas structurally corrected directions give negative inclinations that agree well from site to site and place them in the northern hemisphere. This is consistent with other geological constraints which indicate that the region has been near or at the southern margin of Eurasia since the Paleocene.

In-situ directions at Site 12 give a mean pole position that does not coincide with any part of the established apparent polar wander path for Eurasia in the Tertiary indicating that the magnetization predates the deformation. Such an interpretation places the mean inclination and declination of Site 12 closer to those of the other sites and also gives a pole close to the APW for Eurasia in the Paleocene (Fig. 7). An alternative explanation of in-situ directions is that the site has undergone a substantial clockwise rotation (80–90°) since the end of the Paleocene when the rocks were initially deformed and that this rotation was produced by motion along the North Anatolian Fault. In view of the lack of evidence of similar rotations at the other Paleocene sites which lie both west and east of Site 12, a more likely explanation is that the rocks at all three red bed sites were magnetized at the same time, i.e. prior to deformation.

The overall evidence, therefore, strongly supports a pre-deformation age for the stable magnetization identified at each of the red bed sites. Since the age of these rocks is reported to be

Danian while the deformation is believed to take place in the latest Paleocene (Thanetian), it is apparent that the age of the magnetization is well constrained to between 55 and 65 May (Harland et al., 1982). The structurally corrected declination values at each of the 3 sites (Table 2) show no significant rotation of the sites with respect to Eurasia indicating little or no post deformational rotation (Fig. 9). This provides strong evidence to show that in the study area, the influence of the North Anatolian Fault is negligible and that therefore the rotations of the Jurassic and Cretaceous sites have been produced by the collision associated with the closure of the Vardar Ocean. In addition, geologic studies of the North Anatolian Fault in this part of Anatolia suggest that there is no substantial strike slip motion associated with the fault and a more significant dip slip motion is documented related to extensional tectonics indicated for western Anatolia (Sengör, 1979; Dewey and Sengör, 1979; Sengör et al., 1985).

Because the deformation is linked to the closure of the Vardar Ocean, the red bed magnetization should correspond closely to the APW path for Eurasia in the Paleocene. The pole position for Site 12 fits the APW path well but the poles for the other two sites imply paleolatitudes substantially south of the Eurasian margin (10° N) throughout the Cenozoic (Fig. 7), and are more compatible with the African APW for the early Tertiary (Tauxe et al., 1983). It is difficult to explain these low latitudes since the available geological evidence indicates that the sites were obviously considerably farther north at this time. As the inclinations are less than those predicted, one possible explanation for this disparity is that some flattening of the magnetization vectors has taken place since the magnetization was acquired. Shallowing of magnetic inclination in red beds has been reported elsewhere (Elston and Purucker, 1979; Tauxe and Kent, 1984; Lovlie and Torsvik, 1984). The degree of shallowing of the original inclination is commonly related to grain shape, grain size and the depositional setting of the samples. In a detailed study of both modern fluvial deposits and laboratory redeposited red sediments, Tauxe and Kent (1984) show that the inclination error varies with ambient field and depends on the relative abundance of plate-like and sphere-shaped grains. Inclination errors of 20–25° are found for ambient inclinations of 50° in the fluvial and laboratory sediments. Comparison of the grain sizes from our Paleocene red beds indicates that both Sites 11 and 13 samples have

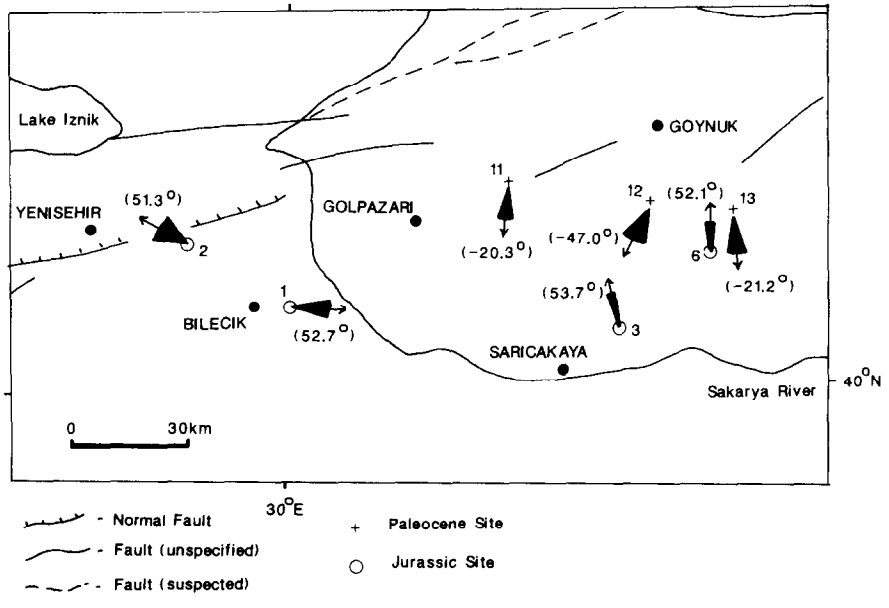


Fig. 9. Enlarged map of study area. Inclinations and declinations shown for the seven sites used in the study. See Fig. 2 for location.

significantly larger grain size than those from Site 12. This is compatible with the magnetic results in which the magnetic inclinations observed at Site 12 are more consistent with the APW of Eurasia and are therefore believed to more accurately reflect the direction of the ambient field than those from the other sites.

Conclusions

The Sakarya Continent has been identified as one of a number of crustal fragments that are intimately involved in the tectonic evolution of the Neo-Tethys. Paleomagnetic results from the Jurassic age sites located on the remains of Sakarya indicate that it was located close to the southern margin of Eurasia throughout much of its history, certainly compatible with the separation of "a few hundred kilometres" suggested by Sengör and Yilmaz (1981). We also suggest that the relative motions of Sakarya tracked those of Eurasia throughout the Mesozoic, i.e. a southerly motion (40° N to 25° N) and a subsequent northerly motion (25° N to 33° N), until in the latest Paleocene, Sakarya docked with Eurasia. It is clear that this collision with Eurasia resulted in the deformation and possible fragmentation of the block as indicated by the different declinations shown by the four Jurassic sites of this study as well as the rotations of other sites reported by other authors. That the rotations in our study were caused by the subsequent activities of the North Anatolian Fault is discounted because Paleocene sites in the vicinity of the rotated Jurassic sites show no evidence of rotations that are related to strike-slip fault activity.

References

Bagin, V.I., Gendler, T.S., Kuz'min, R.N., Rybak, R.S. and Urazayeva, T.K., 1976. The weak ferromagnetism of natural hydrogoethites. Izy. Acad. Sci. U.S.S.R. Phys. Solid Earth, 12: 328–333 (English edn.).

Dewey, J.F. and Sengör, A.M.C., 1979. Aegean and surrounding regions: complex multiplate and continuum tectonics in a convergent zone. Geol. Soc. Am. Bull., 90: 84–92.

Elston, D.P. and Purucker, M.E., 1979. Detrital magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona. J. Geophys. Res., 84: 1653–1665.

Eroskay, S.O., 1965. Geology of the Paslar Gorge–Golpazari area. Istanbul Univ. Fen. Fak. Mecm., Ser. B, 30 (3/4).

Evans, I., Hall, S.A., Carman, M.F., Senalp, M. and Coskun, S., 1982. A paleomagnetic study of the Bilecik Limestone (Jurassic), northwestern Anatolia. Earth Planet. Sci. Lett., 61: 199–208.

Gregor, C.B. and Zijderveld, J.D.A., 1964. The magnetism of some Permian red sandstones from northwestern Turkey. Tectonophysics, 1: 289–306.

Guner, M., 1982. A paleomagnetic study of some basaltoids and ores from the Pontic Ranges, northern Turkey: palaeogeographic implications. Tectonophysics, 90: 309–333.

Gurpinar, O., 1977. Geological investigation of the Bilecik–Inegol–Yenisehir territories together with a study of engineering properties of the Bilecik Limestone. Istanbul Univ. Fen. Fak. Mecm., Ser. B, 40 (1–4): 83–113.

Harland, W.B., Cox, A.V., Llewellyn, P.G., Pickton, C.A.G., Smith, A.G. and Walters, R., 1982. A Geologic Time Scale. Cambridge University Press, Cambridge, 131 pp.

Hedley, I.G., 1971. The weak ferromagnetism of goethite (α-FeOOH). Z. Geophys., 37: 409–420.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc., 62: 699–718.

Kleist, R., Hall, S.A. and Evans, I., 1984. A paleomagnetic study of the Lower Cretaceous Cupido Limestone, northeast Mexico: evidence for local rotation within the Sierra Madre Oriental. Geol. Soc. Am. Bull., 95: 55–60.

Lauer, J.P., 1981. L'évolution géodynamique de la Turquie et de Chypre déduite de l'étude paléomagnetique. Thèse ULP Strasbourg, 299 pp.

Lauer, J.P., 1984. Geodynamic evolution of Turkey and Cyprus based on paleomagnetic data. In: J.E. Dixon and A.H.F. Robertson (Editors), The Geological Evolution of the Eastern Mediterranean. Geol. Soc. London, Spec. Publ., 17: 483–491.

Lovlie, R. and Torsvik, T., 1984. Magnetic remanence and fabric properties of laboratory-deposited hematite bearing red sandstone. Geophys. Res. Lett., 11: 221–224.

Luyendyk, B.P., Kamerling, N.J., Terres, R.R. and Hornafius, J.S., 1985. Simple shear of southern California during Neogene time suggested by paleomagnetic declinations. J. Geophys. Res., 90: 12,445–12,466.

Ogg, J.G. and Lowrie, W., 1986. Magnetostratigraphy of the Jurassic/Cretaceous boundary. Geology, 14: 547–550.

Ogg, J.G., Steiner, M.B., Oloriz, F. and Taverna, J.M., 1984. Jurassic magnetostratigraphy: Kimmeridgian–Tithonian of Sierra Gorda and Carcabuey, southern Spain. Earth Planet. Sci. Lett., 71: 147–162.

Orbay, N., 1979. Paleomagnetism of the North Anatolian Fault Zone. Rev. Fac. Sci. Univ. Istanbul, C., 44: 23–39.

Orbay, N. and Bayburdi, A., 1979. Paleomagnetism of dykes and tuffs from the Mesudiye region and the rotation of Turkey. Geophys. J. R. Astron. Soc., 59: 437–444.

Saner, S., 1980. The paleogeographical interpretation of the

Mudurnu–Goynuk Basin based on the depositional features of the Jurassic and later ages. Geol. Soc. Turk. Bull., 23: 39–52.

Sanver, M., 1968. A paleomagnetic study of Quaternary volcanic rocks from Turkey. Phys. Earth Planet. Inter., 1: 403–421.

Sanver, M. and Ponat, E., 1981. Paleomagnetic results from Kirzehi and surrounding area—rotation of the Kirzehir Massif. Istanbul Earth Sci. Rev., 1: 231–238.

Saribudak, M., Sanver, M. and Ponat, E., 1989. Location of the western Pontides, NW Turkey, during Triassic time: preliminary paleomagnetic results. Geophys. J. R. Astron. Soc., 96: 43–50.

Sengör, A.M.C., 1979. The North Anatolian transform fault: its age, offset and tectonic slgnificance. J. Geol. Soc. London, 136: 269–282.

Sengör, A.M.C. and Yilmaz, Y., 1981. Tethyan evolution of Turkey: a plate tectonic approach. Tectonophysics, 75: 181–241.

Sengör, A.M.C., Yilmaz, Y. and Sungurlu, O., 1984. Tectonics of the Mediterranean Cimmerides: nature and evolution of the western termination of Palaeo-Tethys. In: J.E. Dixon and A.H.F. Robertson, The Geological Evolution of the Eastern Mediterranean. Geol. Soc. London, Spec. Publ., 17: 77–112.

Sengör, A.M.C., Gorur, N. and Saroglu, F., 1985. Strike-slip faulting and related basin formation in zones of tectonic escape: Turkey as a case study. In: K.T. Biddle and N. Christie-Blick (Editors), Strike Slip Deformation, Basin Formation, and Sedimentation. Soc. Econ. Paleontol. Mineral., Spec. Publ., 37: 227–264.

Strangway, D.W., Honea, R.M., McMahon, B.E. and Larson, E.E., 1968. The magnetic properties of naturally occurring goethite. Geophys. J. R. Astron. Soc., 15: 345–359.

Tauxe, L. and Kent, D.V., 1984. Properties of a detrital remanence carried by haematite from the study of modern river deposits and laboratory redeposition experiments. Geophys. J. R. Astron. Soc., 76: 543–561.

Tauxe, L., Besse, J. and LaBreque, J.L., 1983. Paleolatitudes from DSDP Leg 73 sediment cores: implications for the apparent polar wander path for Africa during the Late Mesozoic and Cenozoic. Geophys. J. R. Astron. Soc., 73: 315–324.

Van der Voo, R., 1968. Jurassic, Cretaceous and Eocene pole positions from northeastern Turkey. Tectonophysics, 6: 251–269.

Van der Voo, R. and Channell, J.E.T., 1980. Paleomagnetism in orogenic belts. Rev. Geophys. Space Phys., 18: 455–482.

Westphal, M., Bazhenov, M.L., Lauer, J.P., Pechersky, D.M. and Sibuet, J.C., 1986. Paleomagnetic implications on the evolution of the Tethys Belt from the Atlantic Ocean to the Pamirs since the Triassic. Tectonophysics, 123: 37–82.